

Informatica PowerCenter

Architecture & ETL — In-Depth Study Notes

NiC IT Academy · Data Warehousing & ETL Architecture

01 · What Is Informatica PowerCenter?

Informatica PowerCenter is an enterprise-grade **ETL (Extract, Transform, Load)** platform used to move and integrate data from heterogeneous source systems into a centralised target such as a data warehouse or data mart. It separates *design-time* (metadata definition via client tools) from *run-time* (data movement via the server engine), making it scalable and manageable across large organisations.

Key concept: Informatica never stores data permanently — it acts as a pipeline. All persistent storage belongs to your source and target databases. Informatica's repository only holds *metadata* (the instructions), not the actual data rows.

02 · Architecture Overview — Four Core Components

The Informatica architecture has four tightly coupled layers. Understanding how they interact is essential for both development and troubleshooting.

Client Tools (R·D·W·M)	Developer PC (Windows)	Design mappings, schedule workflows, monitor execution
Integration Service	Informatica Server (Unix/Linux)	Core ETL engine — reads, transforms, and writes data
Repository Service	Informatica Server (Unix/Linux)	Manages metadata access, versioning, and client connections
Repository Database	RDBMS e.g. Oracle	Stores all metadata: mappings, workflows, sessions, configs

03 · Client Tools — R · D · W · M

All four client tools are installed on the ETL developer's Windows workstation and communicate with the Informatica Server over TCP/IP. They interact only with metadata — they never touch source or target data directly.

R **Repository Manager**
Creates, deletes, copies, and backs up repositories. Manages folder structures, user accounts, permissions, and locks. Used by administrators, not day-to-day developers.

D **Designer**
The primary development environment. Used to: (1) import source definitions from databases or flat files; (2) define target schemas; (3) build Mappings by connecting source qualifiers through transformation objects to targets. Mappings are the blueprints the Integration Service executes.

W **Workflow Manager**
Wraps mappings inside Sessions (which hold runtime settings like connection info, commit intervals, error handling). Sessions are linked into Workflows with scheduling, event triggers, email notifications, and decision logic. You cannot run a mapping without a workflow.

M **Workflow Monitor**
A real-time and historical dashboard showing: workflow/session run status, start/end times, rows read/written/rejected, and transformation-level statistics. Indispensable for debugging failed loads.

04 · Integration Service — The ETL Engine

The Integration Service is the only component that physically moves data. It runs on Unix/Linux and is invoked when the Workflow Manager triggers a session. Internally it executes a three-stage pipeline:

Stage	Operation	Detail
Extract (Reader)	SQL SELECT	Connects to source via ODBC/native drivers. Issues SQL generated from the Source Qualifier transformation. Reads data in blocks called 'DTM buffers' for efficiency.
Transform	Cleanse / Enrich	Applies mapping logic: Filter, Joiner, Aggregator, Lookup, Expression, Router, Sequence Generator, Sorter, Update Strategy, and more. Runs entirely in memory (or via cache files for large lookups/aggregations).
Load (Writer)	INSERT / UPDATE	Writes processed rows to the target. The Update Strategy transformation flags each row as insert, update, delete, or reject. Commits occur at the configured commit interval.

Pipeline architecture: Data flows record-by-record (or buffer-by-buffer) through Reader → Transformation → Writer without being written to disk between stages — unless you use a persistent lookup cache or the Sorter transformation, which may spill to disk for very large datasets.

05 · Repository Service

The Repository Service is a daemon process on the Informatica Server that acts as the gatekeeper between all other components and the Repository Database. It ensures:

- **Concurrent access:** Multiple designers can connect and work simultaneously without corrupting metadata.

- **Object locking:** When a developer opens a mapping for editing, it is locked so no one else can modify it at the same time.
- **Version control:** In versioned mode, every save creates a new version of the object, enabling rollback.
- **Security:** Enforces folder-level and object-level permissions for different user groups.
- **Runtime metadata delivery:** When the Integration Service starts a session, it contacts the Repository Service to retrieve the mapping instructions — not the Repository DB directly.

06 · Repository Database

The Repository Database is a standard relational database (Oracle is most common; SQL Server and DB2 are also supported). It stores every object created by the client tools:

Object	What is stored
Mappings	Source/target definitions and all transformation logic
Sessions	Runtime configurations: connections, commit size, error thresholds
Workflows	Scheduling, event triggers, task sequences, decision logic
Transformations	Reusable transformation objects shared across mappings
Connections	Database connection strings (encrypted credentials)
Metadata extensions	Custom attributes added to any Informatica object

07 · End-to-End ETL Flow Summary

The diagram below summarises how all components interact during a single workflow run:

1 Developer (Client Tools)

Designer creates mapping. Workflow Manager wraps it in a session and schedules a workflow.

2 Repository Service

Stores all metadata in the Repository DB. Delivers mapping instructions to the Integration Service at runtime.

3 Integration Service — Reader

Connects to source DB. Runs SQL SELECT. Reads rows into DTM memory buffers.

4 Integration Service — Transform

Applies all transformation logic defined in the mapping. Filters, joins, aggregates, lookups, expressions.

5 Integration Service — Writer

Commits processed rows to the target DB via INSERT / UPDATE. Rejects bad rows to a reject file.

6 Workflow Monitor (Client Tools)

Developer monitors run status, row counts, and errors in real time.

08 · Key Relationships & Exam Tips

Design-time vs run-time: Client Tools + Repository Service = design-time (no data moves). Integration Service = run-time (actual data movement). This separation is Informatica's core architectural principle.

Clients never touch the Repository DB directly: All client tool requests go through the Repository Service. Direct DB connections are reserved for the Repository Service itself and DBA backup tasks.

Integration Service is stateless: It reads instructions from the Repository Service each time a session runs. No mapping logic is cached on the server — everything lives in the repository.

One Integration Service, many workflows: A single Integration Service can run multiple workflows concurrently (subject to DTM buffer memory limits). Load balancing across multiple Integration Services is also supported.

Unix/Linux requirement for the server: The Informatica Server (Integration Service + Repository Service) requires Unix/Linux. Only the Client Tools run on Windows. This is a common exam question.

09 · Common Transformation Types

Transformations are the building blocks inside a mapping. They fall into two categories:

Transformation	Type	Purpose
Source Qualifier	Active / Connected	Generates the SQL query to extract data from the source
Filter	Active / Connected	Removes rows that do not meet a condition
Expression	Passive / Connected	Computes new values, type conversions, string manipulation
Aggregator	Active / Connected	Performs GROUP BY operations: SUM, COUNT, AVG, MAX, MIN
Joiner	Active / Connected	Joins two heterogeneous data sources (like a SQL JOIN)
Lookup	Passive / Connected or Unconnected	Looks up values in a table or flat file; supports cache
Router	Active / Connected	Routes rows to different target groups based on conditions
Update Strategy	Active / Connected	Flags each row: DD_INSERT, DD_UPDATE, DD_DELETE, DD_REJECT
Sequence Generator	Passive / Connected	Generates unique numeric keys (surrogate keys)
Sorter	Active / Connected	Sorts rows; may use disk cache for large datasets

